

TARGET-90

JKBOSE CLASS-11TH

STRAIGHT LINES

One Shot

MATHS

Mehraj sir



JKBOSE WALLAH

Topics *to be covered*



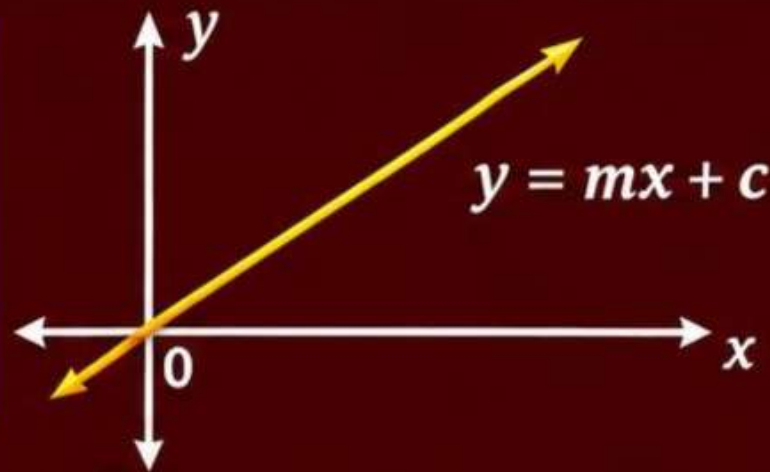
Slope of a Line

STRAIGHT LINES



1 Definition

A straight line is a path formed by a point moving in a fixed direction.



2 Equation of a Straight Line

$$y = mx + c$$

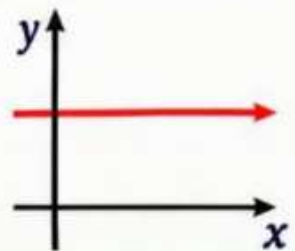
m → slope (gradient)

c → y-intercept (where line cuts y-axis)

x, y → coordinates

3 Types of Straight Lines

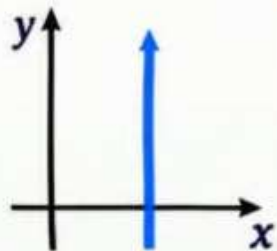
Horizontal Line



$$y = c$$

(slope = 0)

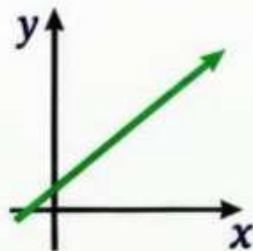
Vertical Line



$$x = a$$

(slope undefined)

Oblique Line



$$y = mx + c$$

($m \neq 0$)

4 Forms of Equation

1. Slope – Intercept Form

$$y = mx + c$$

2. Point – Slope Form

$$y - y_1 = m(x - x_1)$$

3. Two – Point Form

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

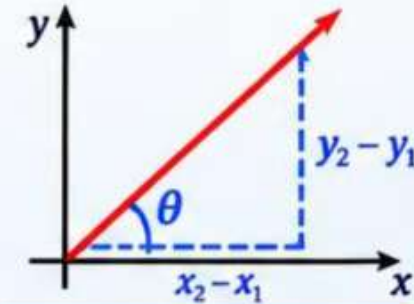
4. Intercept Form

$$\frac{x}{a} + \frac{y}{b} = 1 \quad (a, b \text{ are intercepts})$$

5 Gradient (Slope)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$



6 Parallel and Perpendicular Lines

Parallel Lines

$$m_1 = m_2$$

e.g. $y = 2x + 3$
 $y = 2x - 1$

Perpendicular Lines

$$m_1 \times m_2 = -1$$

e.g. $y = 3x + 2$
 $y = -\frac{1}{3}x + 5$

7 Key Points to Remember



Slope tells the inclination of the line.



Parallel lines have the same slope.



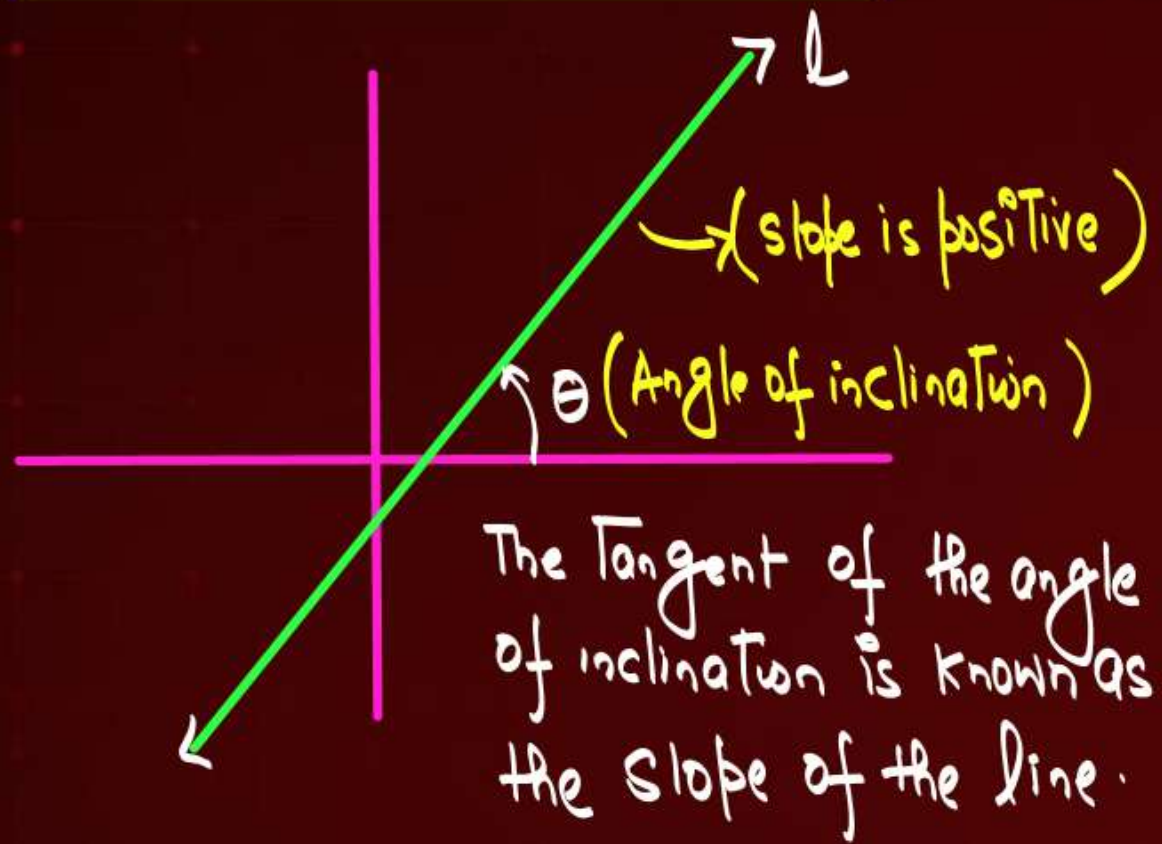
Perpendicular lines have slopes whose product is -1 .



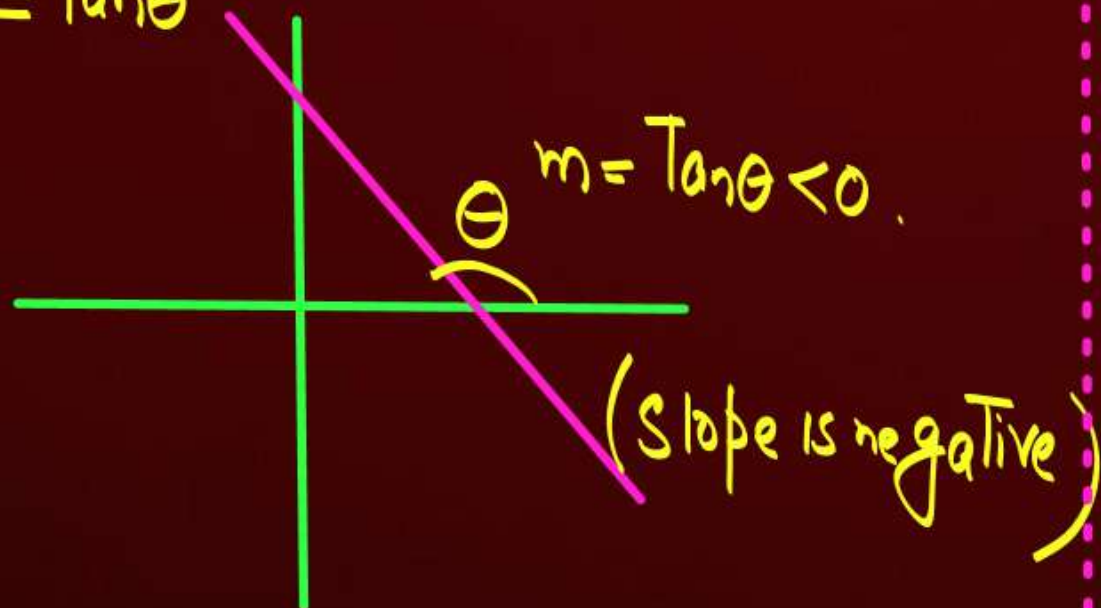
A straight line can be represented in many forms of equation.



Slope of a Line



$$m = \tan \theta$$



★ Slope of a line passing through two points:



The slope of the line passing through $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by;

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Angle Between Two Lines



Suppose that θ be the acute angle b/w l_1 and l_2 , then

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

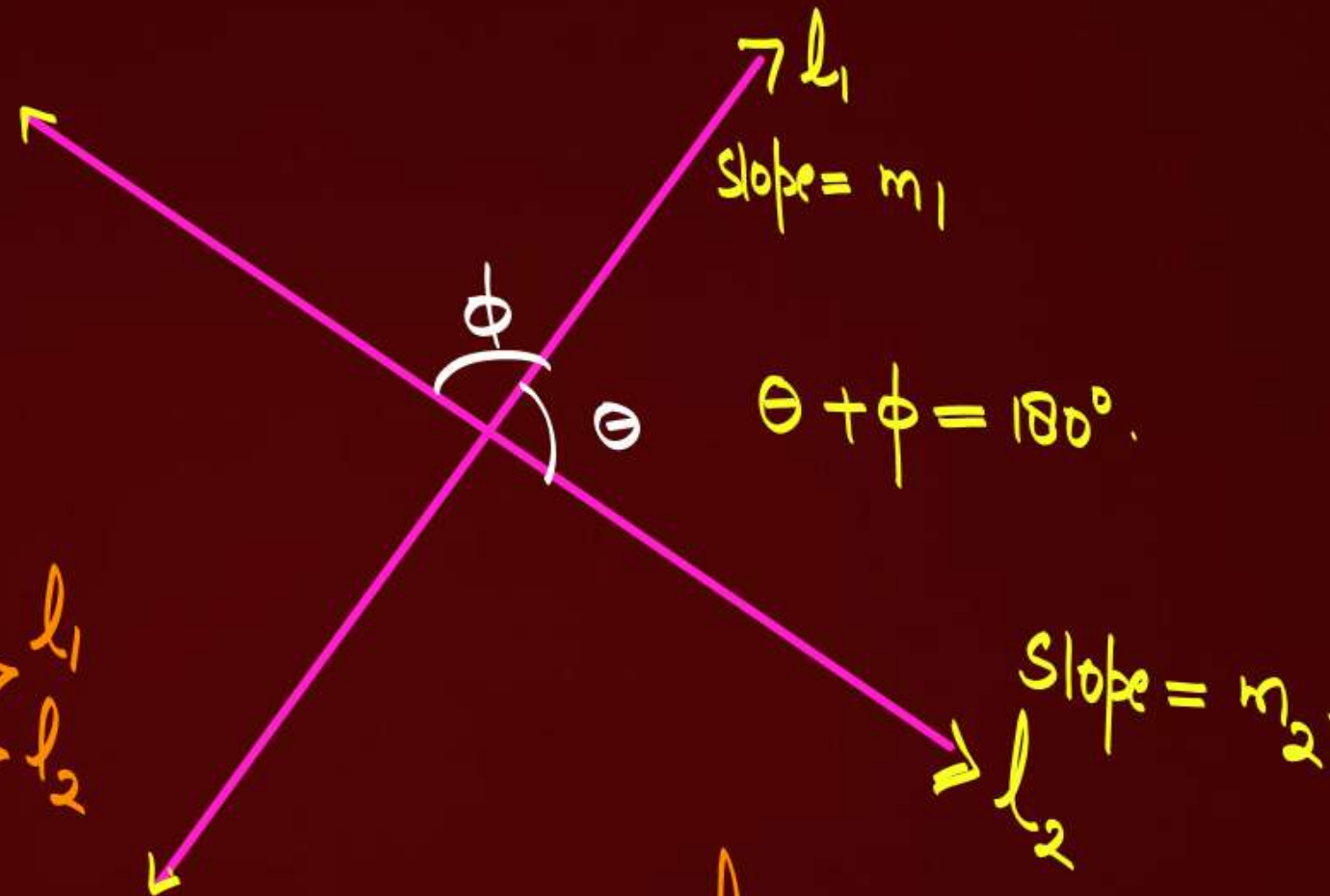
* If l_1 is parallel to l_2 , then $\theta = 0$

$$\tan \theta = 0$$

$$m_2 - m_1 = 0$$

$$\Rightarrow m_1 = m_2$$

$\Rightarrow$ For parallel lines, the slopes are equal.



* For perpendicular line, $\theta = \frac{\pi}{2}$
 $1 + m_1 m_2 = 0$

$$\Rightarrow m_1 m_2 = -1$$

$\Rightarrow$ for perpendicular lines, the product of Slopes is -1 .

EXAMPLE

If the angle between two lines is $\frac{\pi}{4}$ and slope of one of the lines is $\frac{1}{2}$, find the slope of the other line.

(Sol) We know that $\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$

$$\Rightarrow \tan\left(\frac{\pi}{4}\right) = \left| \frac{m - \frac{1}{2}}{1 + \frac{1}{2}m} \right| = \left| \frac{2m-1}{2+m} \right|$$

$$\Rightarrow 1 = \left| \frac{2m-1}{m+2} \right|$$

$$\Rightarrow \frac{2m-1}{m+2} = \pm 1$$

(Important)

$$\Rightarrow 2m-1 = \pm(m+2)$$

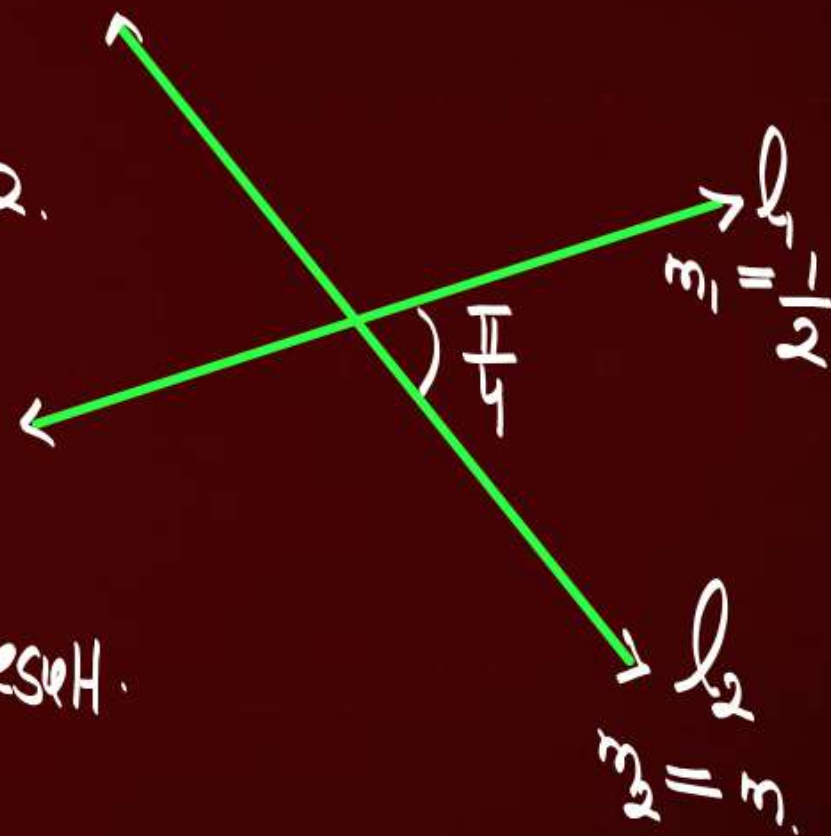
$$\Rightarrow 2m-1 = m+2, -m-2$$

$$\Rightarrow 2m-m = 2+1$$

$$\Rightarrow 2m+m = -2+1$$

$$\Rightarrow m = 3, -\frac{1}{3}$$

Hence the result.



EXAMPLE



Line through the points $(-2, 6)$ and $(4, 8)$ is perpendicular to the line through the points $(8, 12)$ and $(x, 24)$. Find the value of x .

(Sol) For perpendicular lines, the product of slopes is -1 .

$$\text{So, } m_{AB} \times m_{CD} = -1$$

$$\Rightarrow \left(\frac{8-6}{4+2} \right) \left(\frac{24-12}{x-8} \right) = -1$$

$$\Rightarrow \frac{2}{\cancel{6}} \times \frac{\cancel{12}^2}{x-8} = -1$$

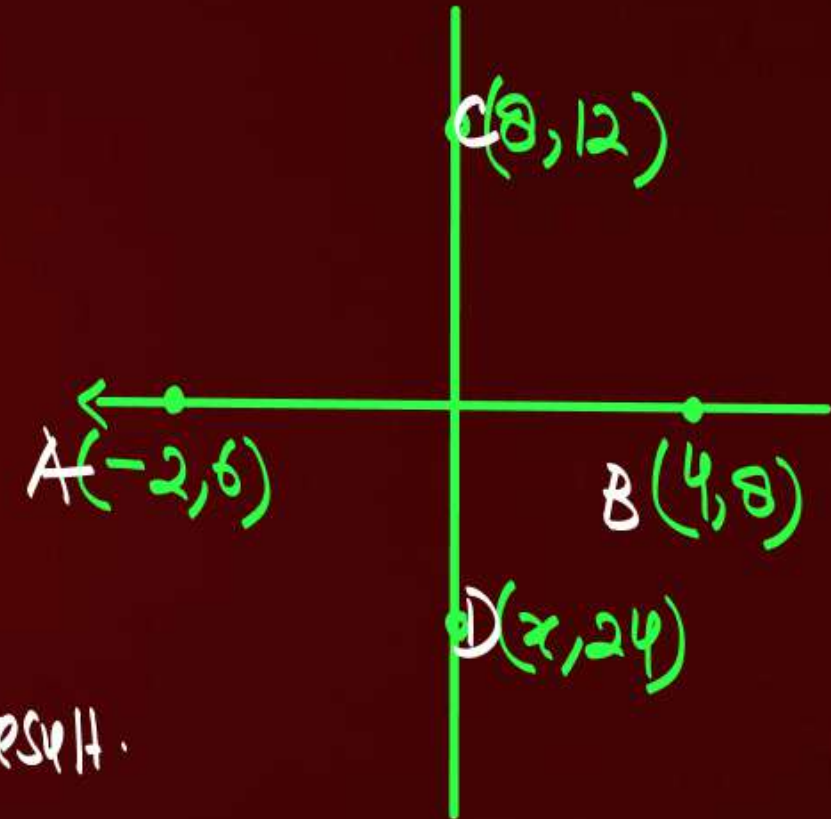
$$\Rightarrow \frac{4}{x-8} = -1$$

$$\Rightarrow 4 = -x + 8$$

$$\Rightarrow x = 8 - 4$$

$$\Rightarrow \boxed{x = 4}$$

Hence the result.



QUESTION

If the line segments AB and CD are parallel, then the value of k is:

1 $2 \times$

2 $4 \checkmark$

3 $8 \times$

4 $3 \times$

$$m_{AB} = m_{CD}$$

$$\frac{2-0}{0-3} = \frac{k-0}{0-6}$$

$$\Rightarrow \frac{2}{-3} = \frac{k}{-6}$$

$$\Rightarrow k = 4$$



QUESTION



$$\begin{array}{c} d \\ \text{---} \\ (x_1, y_1) \quad (x_2, y_2) \end{array} \quad d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Find a point on the **x-axis** which is equidistant from the points **(7, 6)** and **(3, 4)**.

(Sol) Suppose that $P(a, 0)$ be the required point which is equidistant from the points $A(7, 6)$ and $B(3, 4)$.

So, $AP = BP$

or ~~$\sqrt{(a-7)^2 + 36} = \sqrt{(a-3)^2 + 16}$~~

~~$\Rightarrow a^2 + 49 - 14a + 36 = a^2 + 9 - 6a + 16$~~

$\Rightarrow -14a + 6a = 25 - 85$

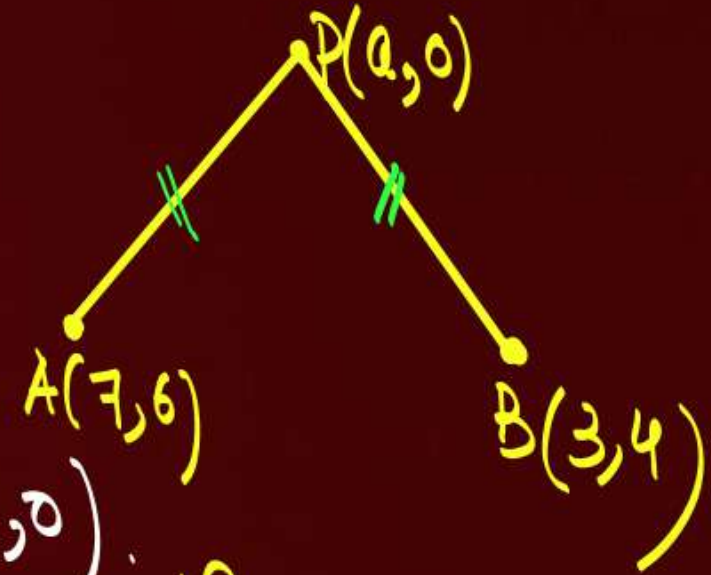
$\Rightarrow -8a = -60$

$\Rightarrow a = \frac{-60}{-8} = \frac{15}{2}$

$\Rightarrow a = \frac{15}{2}$

So, the required point on the x-axis is $P(\frac{15}{2}, 0)$.

Hence the result.



QUESTION

Find the slope of a line, which passes through the origin, and the mid-point of the line segment joining the points $P(0, -4)$ and $B(8, 0)$.

H/W

QUESTION



Find the angle between the x -axis and the line joining the points $(3, -1)$ and $(4, -2)$.

h/w

QUESTION



(Important Sawall)

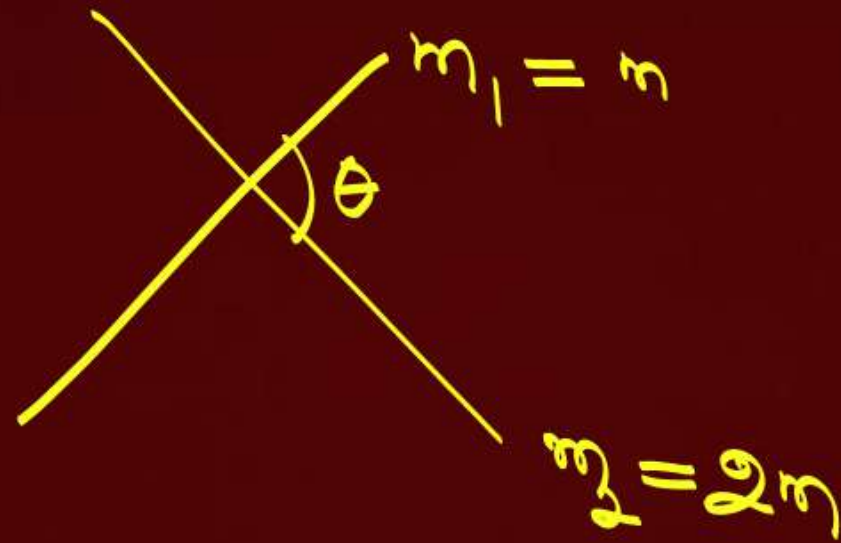
The slope of a line is double of the slope of another line. If tangent of the angle between them is $1/3$, find the slopes of the lines.

$$\tan \theta = \frac{1}{3}$$

$$\Rightarrow \left| \frac{2m - m}{1 + 2m \times m} \right| = \frac{1}{3}$$

$$\Rightarrow \frac{m}{1 + 2m^2} = \frac{1}{3} \text{ or } -\frac{1}{3}$$

$$\frac{m}{1 + 2m^2} = \frac{1}{3}, \quad \frac{m}{1 + 2m^2} = -\frac{1}{3}$$



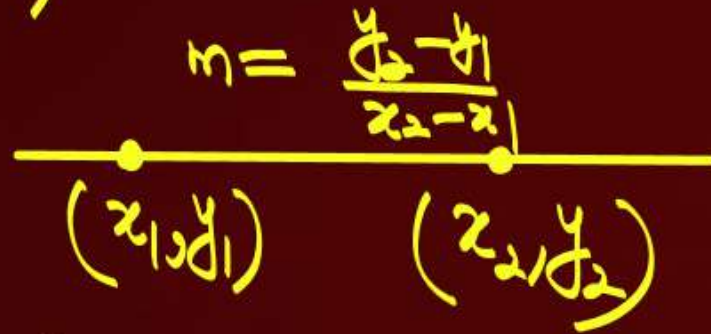
$$\frac{m}{1 + 2m^2} = \frac{1}{3}$$
$$\frac{m}{1 + 2m^2} = -\frac{1}{3}$$

Various Forms of the Equation of a Line

Point-slope form:


$$y - y_1 = m(x - x_1)$$

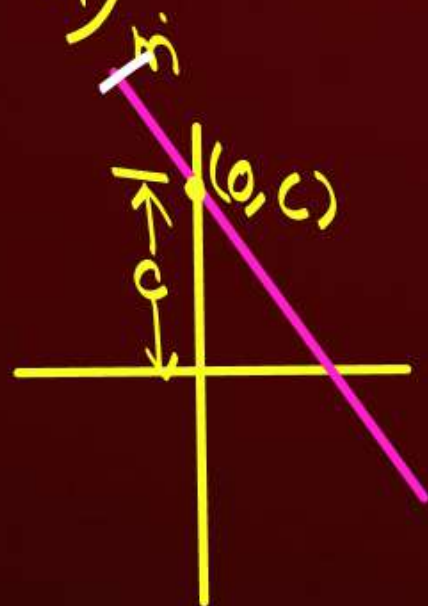
Two-point form:


$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$$

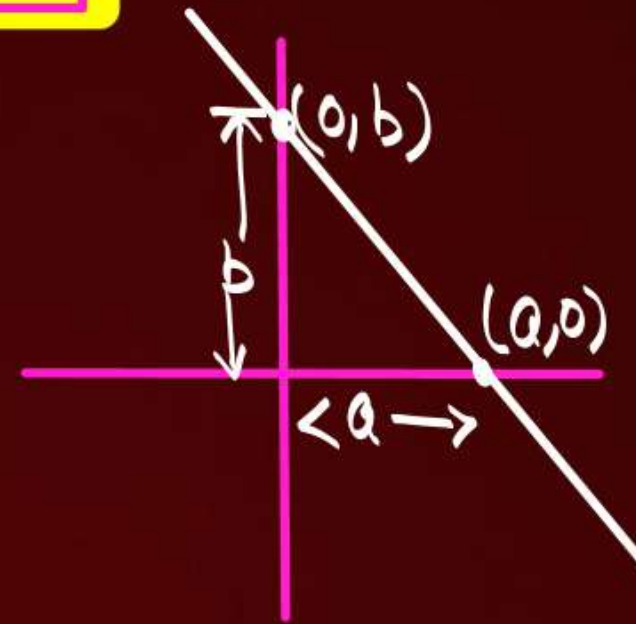
Slope-intercept form:

$$y = mx + c$$



Intercept form:

$$\frac{x}{a} + \frac{y}{b} = 1$$



General form:

$$ax + by + c = 0 \quad \text{--- (1)}$$

$$\text{Slope of (1)} = -\frac{a}{b} = -\left(\frac{\text{Co. ff. of } x}{\text{Co. ff. of } y} \right)$$

EXAMPLE



Find the equation of the line through $(-2, 3)$ with slope -4 .

$$y - 3 = -4(x + 2)$$

$$\Rightarrow y - 3 = -4x - 8$$

$$\Rightarrow 4x + y + 5 = 0.$$



EXAMPLE

Find the equation of the Perpendicular bisector of the line segment joining the points A(2,3) and B(6, -5).

(Sol) Here by Mid-point formula,

$$M(x, y) = M\left(\frac{2+6}{2}, \frac{3-5}{2}\right)$$

$$\Rightarrow M(x, y) = M(4, -1)$$

Also, we know that $m \times m_{AB} = -1$.

$$\Rightarrow m\left(\frac{-5-3}{6-2}\right) = -1.$$

$$\Rightarrow m\left(\frac{-8}{4}\right) = -1 \Rightarrow m = \frac{1}{2}$$

Therefore the required equation is:

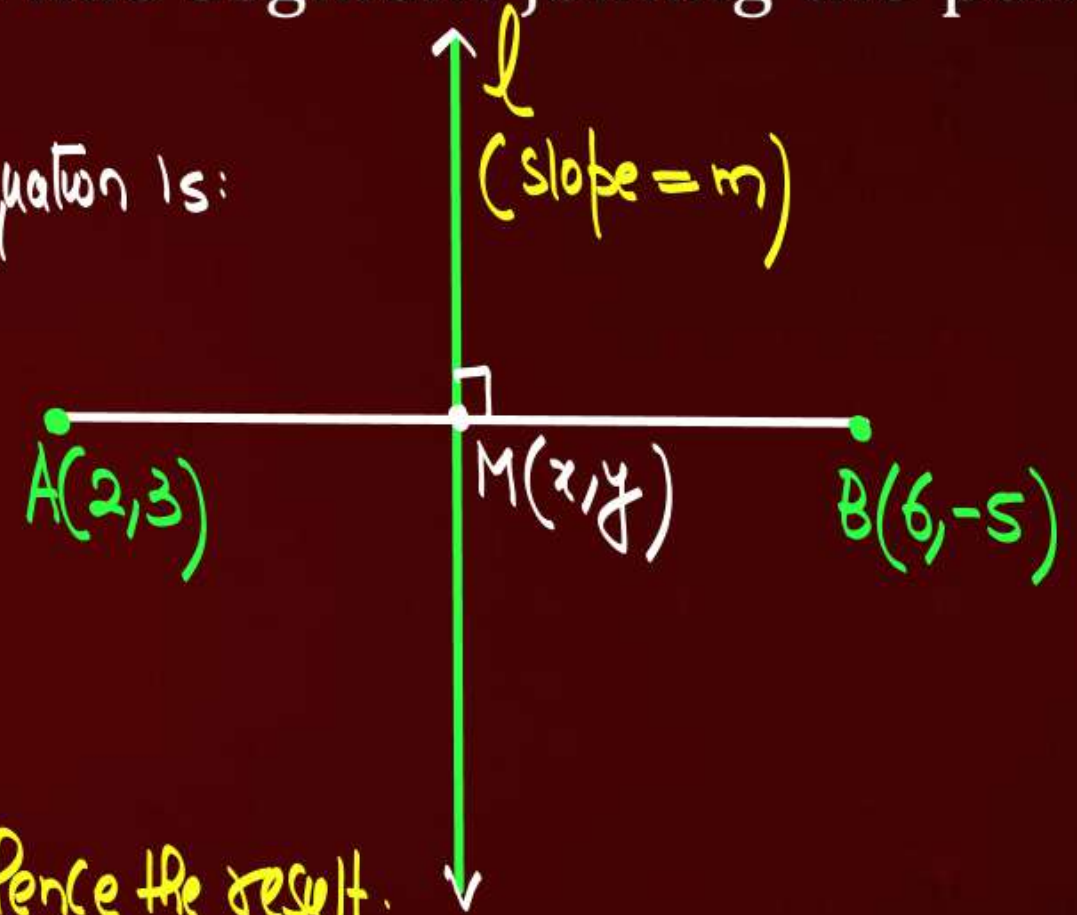
$$y - y_1 = m(x - x_1)$$

$$\Rightarrow y + 1 = \frac{1}{2}(x - 4)$$

$$\Rightarrow 2y + 2 = x - 4$$

$$\Rightarrow \boxed{x - 2y - 6 = 0}$$

Hence the result.



QUESTION (NCERT EXAMPLAR):

The inclination of the line $x - y + 3 = 0$ with the positive direction of x -axis is.

1 135°

2 45°

3 -45°

4 -135°

QUESTION (NCERT EXAMPLAR):

The two lines $ax + by = c$ and $a'x + b'y = c'$ are perpendicular if.

1 ✓ $aa' + bb' = 0$

2 ✗ $ab' = ba'$

3 ✗ $ab + a'b' = 0$

4 ✗ $ab' + ba' = 0$

$$-\frac{a}{b} \times -\frac{a'}{b'} = -1.$$

$$\Rightarrow \frac{aa'}{bb'} = -1$$

$$\Rightarrow aa' = -bb'$$

$$\Rightarrow aa' + bb' = 0.$$

QUESTION (NCERT EXAMPLAR):

For specifying a straight line, show many geometrical parameters should be known?

- 1 ~~1~~
- 2 ✓✓
- 3 ~~4~~
- 4 ~~3~~

QUESTION



In Exercises 1 to 8, find the equation of the line which satisfy the given conditions:

Passing through $(2, 2\sqrt{3})$ and inclined with the x-axis at an angle of 75° .

(Sol) Here $m = \tan \theta = \tan 75^\circ$

$$\Rightarrow m = \tan(45^\circ + 30^\circ) = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$$

$$\Rightarrow m = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

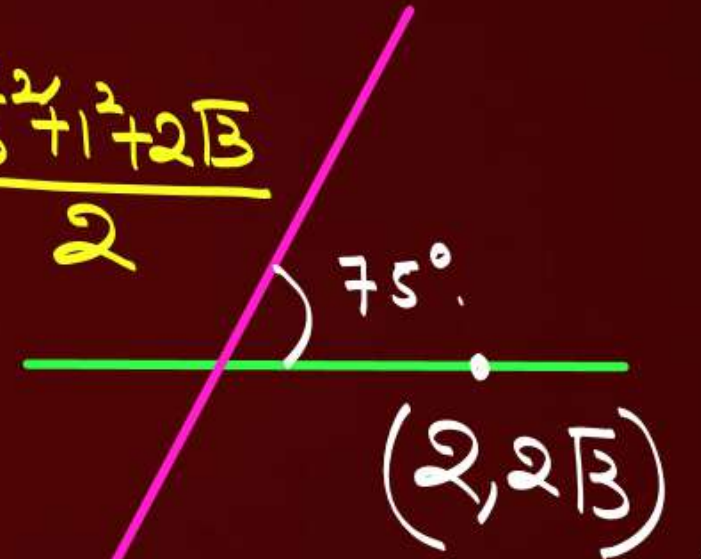
$$\Rightarrow m = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$\Rightarrow m = \frac{(\sqrt{3} + 1)^2}{\sqrt{3}^2 - 1^2} = \frac{\sqrt{3}^2 + 1^2 + 2\sqrt{3}}{2}$$

$$\Rightarrow m = \frac{4 + 2\sqrt{3}}{2}$$

$$\Rightarrow m = \frac{2(2 + \sqrt{3})}{2}$$

$$\Rightarrow m = 2 + \sqrt{3}$$



Therefore, the required equation of the line is:

$$y - 2\sqrt{3} = (2 + \sqrt{3})(x - 2)$$

$$\Rightarrow y - 2\sqrt{3} = (2 + \sqrt{3})x - 4 - 2\sqrt{3}$$

$$\Rightarrow (2 + \sqrt{3})x - y - 4 = 0$$

Hence the result.

QUESTION

In Exercises 1 to 8, find the equation of the line which satisfy the given conditions:

The vertices of ΔPQR are $P (2, 1)$, $Q (-2, 3)$ and $R (4, 5)$. Find equation of the median through the vertex R . *h/w*

QUESTION



Find the equation of the line passing through $(-3, 5)$ and perpendicular to the line through the points $(2, 5)$ and $(-3, 6)$.

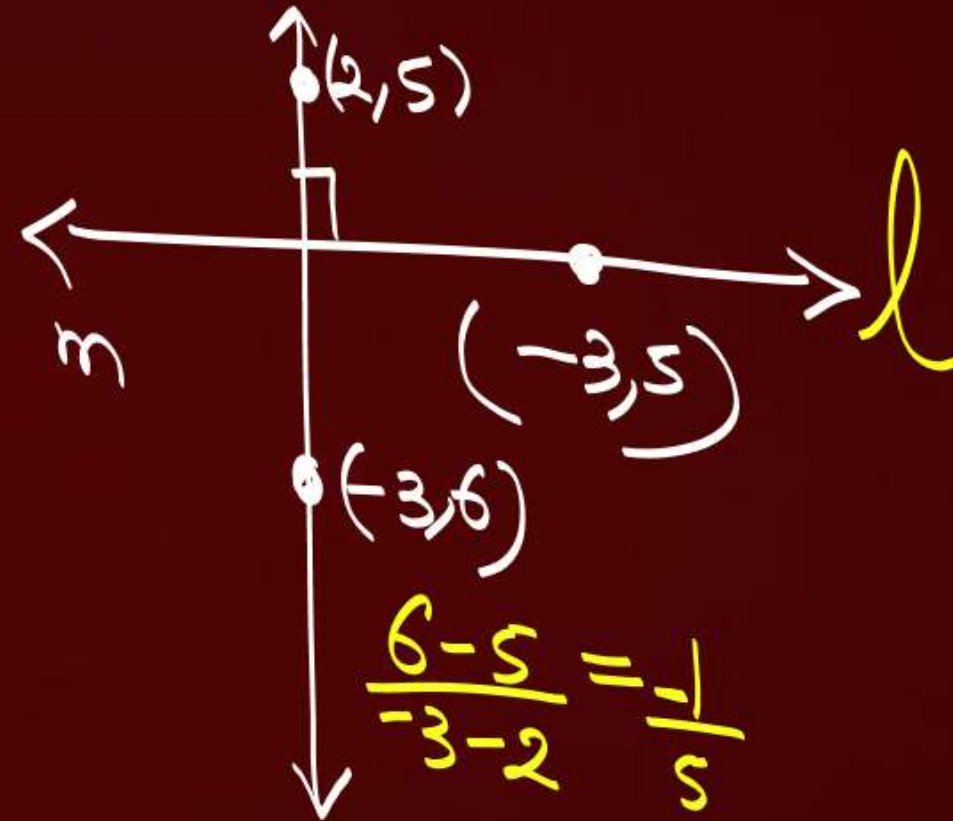
$$-\frac{1}{5} \times m = -1$$

$$\Rightarrow m = 5$$

$$y - 5 = 5(x + 3)$$

$$\Rightarrow 5x + 15 = y - 5$$

$$\Rightarrow 5x - y + 20 = 0$$



QUESTION

Find the equation of a line that cuts off equal intercepts on the coordinate axes and passes through the point $(2, 3)$.

h/w.

QUESTION



(Important Sawal)

Find equation of the line passing through the point $(2, 2)$ and cutting off intercepts on the axes whose sum is 9.

(Sol) Suppose that the required equation is:

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \text{--- (1)}, \text{ where } a \text{ \& } b$$

are respectively the intercepts on x -axis
& y -axis.

$$\text{Here } a + b = 9$$

$$\text{or } b = 9 - a \quad \text{--- (2)}$$

Also, Since $(2, 2)$ lies on (1), therefore

$$\frac{2}{a} + \frac{2}{b} = 1$$

$$\Rightarrow 2b + 2a = ab \quad \text{--- (3)}$$

Using (2) in (3), we get:

$$2(9 - a) + 2a = a(9 - a)$$

$$\Rightarrow 18 - \cancel{2a} + \cancel{2a} = 9a - a^2$$

$$\Rightarrow a^2 - 9a + 18 = 0$$

$$\Rightarrow (a - 6)(a - 3) = 0$$

$$\Rightarrow a = 6, 3$$

$$\text{So } b = 9 - a = 9 - 6, 9 - 3$$

$$\Rightarrow b = 3, 6$$

Therefore the required equations are:

$$\frac{x}{6} + \frac{y}{3} = 1; \quad \frac{x}{3} + \frac{y}{6} = 1$$

$$\Rightarrow 3x + 6y = 18, \quad 6x + 3y = 18$$

$$\Rightarrow x + 2y = 6, \quad 2x + y = 6$$

Hence the result.

QUESTION



By using the concept of equation of a line, prove that the three points $(3, 0)$, $(-2, -2)$ and $(8, 2)$ are collinear.

(Sol) The equation of the line passing through $(3, 0)$ and $(8, 2)$ is:

$$y - 0 = \left(\frac{2 - 0}{8 - 3} \right) (x - 3)$$

$$\Rightarrow y = \frac{2}{5} (x - 3)$$

$$\Rightarrow 5y = 2x - 6$$

$$\Rightarrow 2x - 5y - 6 = 0 \quad \text{--- (1)}$$

Here clearly $(-2, -2)$ satisfies

& hence the points $(3, 0)$

, $(-2, -2)$ and $(8, 2)$ are collinear.



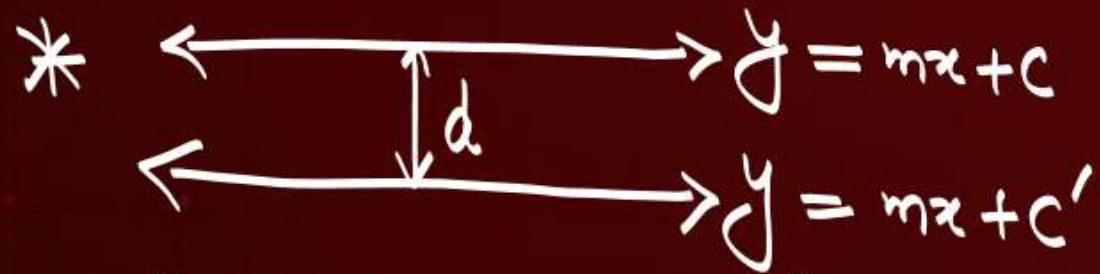
Hence proved

Distance of a Point From a Line



$$d = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

$$ax + by + c = 0$$

* 

Distance b/w parallel lines

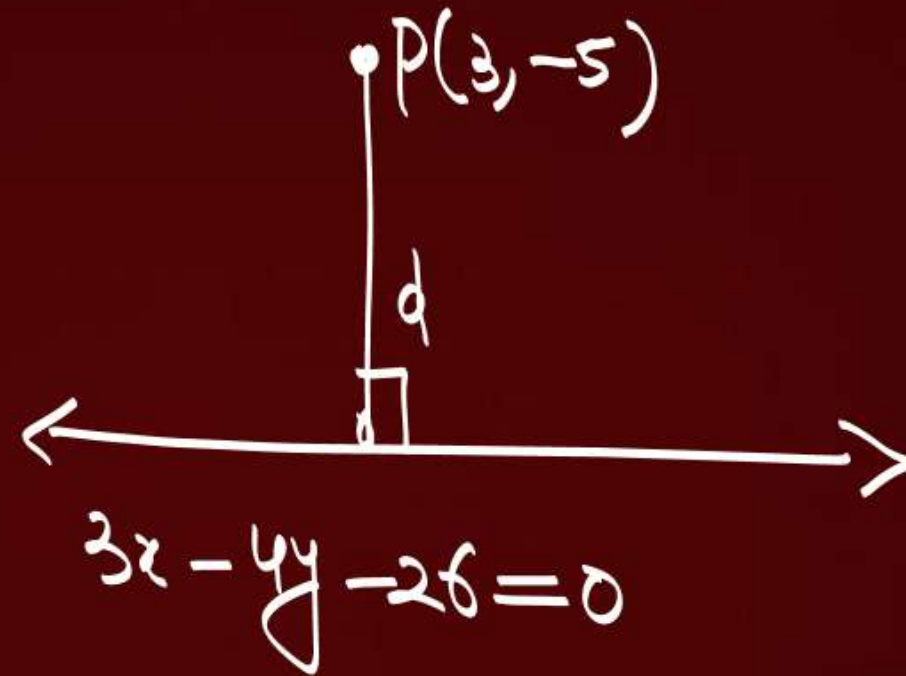
$$= \frac{|c' - c|}{\sqrt{1 + m^2}}$$

EXAMPLE

Find the distance of the point $(3, -5)$ from the line $3x - 4y - 26 = 0$.

(Sol) Here $d = \frac{|3 \times 3 - 4 \times -5 - 26|}{\sqrt{9 + 16}}$

$$\Rightarrow d = \frac{|9 + 20 - 26|}{5} = \frac{3}{5} \text{ units}$$



QUESTION (NCERT EXAMPLAR):

The distance of the point $P(1, -3)$ from the line $2y - 3x = 4$ is

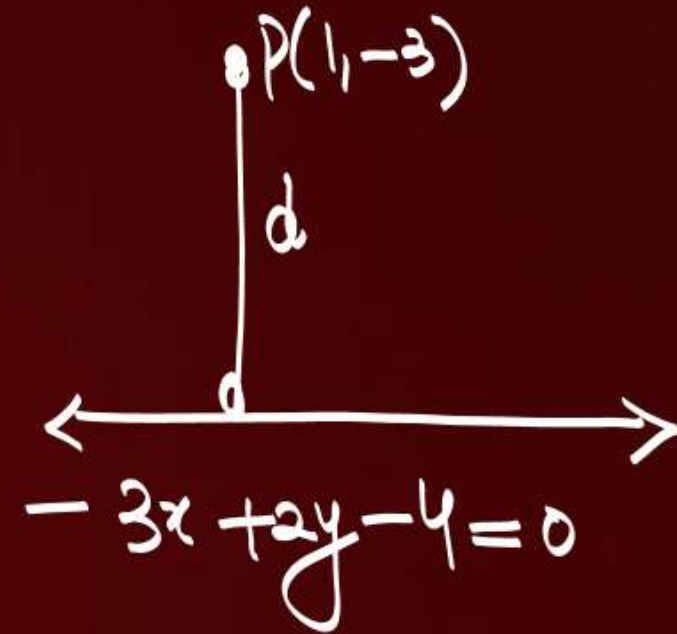
- 1 13 ✗
- 2 $\frac{7}{13}\sqrt{13}$ ✗
- 3 $\sqrt{13}$ ✓
- 4 None of these ✗

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} = \frac{|-3(1) + 2(-3) - 4|}{\sqrt{9 + 4}}$$

$$d = \frac{|-3 - 6 - 4|}{\sqrt{13}} = \frac{|-13|}{\sqrt{13}}$$

$$\Rightarrow d = \frac{13\sqrt{13}}{\sqrt{13}^2} = \frac{13\sqrt{13}}{13}$$

$$d = \sqrt{13}$$



EXAMPLE

Find the distance between the parallel lines $3x - 4y + 7 = 0$ and $3x - 4y + 5 = 0$.

$$\begin{array}{lcl} y = \frac{3}{4}x + \frac{5}{4} & \xleftarrow{\hspace{1cm}} & 3x - 4y + 5 = 0 \\ y = \frac{3}{4}x + \frac{7}{4} & \xleftarrow{\hspace{1cm}} & 3x - 4y + 7 = 0 \end{array}$$

$y = mx + c$
 $d = \frac{|c' - c|}{\sqrt{1 + m^2}}$

$$\begin{aligned} d &= \frac{\left| \frac{7}{4} - \frac{5}{4} \right|}{\sqrt{1 + \frac{9}{16}}} = \frac{\left(\frac{1}{2} \right)}{\sqrt{\frac{25}{16}}} = \frac{\left(\frac{1}{2} \right)}{\left(\frac{5}{4} \right)} = \frac{1}{2} \times \frac{4}{5} \\ &= \frac{2}{5} \text{ units.} \end{aligned}$$

QUESTION (NCERT EXAMPLAR):

The distance between the lines $y = mx + c_1$ and $y = mx + c_2$ is

- 1 $\frac{c_1 - c_2}{\sqrt{m^2 + 1}}$ ✗
- 2 $\frac{|c_1 - c_2|}{\sqrt{1 + m^2}}$ ✓
- 3 $\frac{c_2 - c_1}{\sqrt{1 + m^2}}$ ✗
- 4 0

$$\begin{array}{l} |c_1 - c_2| \\ |c_2 - c_1| \end{array}$$

QUESTION

Reduce the following equations into slope – intercept form and find their slopes and the y -intercepts.

(i) $x + 7y = 0$

(ii) $6x + 3y - 5 = 0$ $\xrightarrow{y = mx + c}$ $3y = -6x + 5$

(iii) $y = 0 = 0 \cdot x + 0$

$y = -2x + \frac{5}{3}$

QUESTION

Reduce the following equations into intercept form and find their intercepts on the axes.

$$\frac{x}{a} + \frac{y}{b} = 1$$

(i) $3x + 2y - 12 = 0$

(ii) $4x - 3y = 6$

(iii) $3y + 2 = 0$

$$3x + 2y = 12$$

$$\frac{3x}{12} + \frac{2y}{12} = 1$$

$$\frac{x}{4} + \frac{y}{6} = 1$$

QUESTION



Find the distance of the point $(-1, 1)$ from the line $12(x + 6) = 5(y - 2)$.

QUESTION



(Important Sawaal)

$$4x + 3y - 12 = 0$$

Find the points on the x -axis, whose distances from the line $\frac{x}{3} + \frac{y}{4} = 1$ are 4 units.

(Sol) Given line is:

$$\frac{x}{3} + \frac{y}{4} = 1$$

$$\Rightarrow 4x + 3y - 12 = 0 \text{ --- (1)}$$

Suppose that $P(a, 0)$ be the required point on the x -axis which is at a distance of 4

units from (1). So,

$$\frac{|4a + 3(0) - 12|}{\sqrt{4^2 + 3^2}} = 4$$

$$\Rightarrow \frac{|4a - 12|}{5} = 4$$

$$\Rightarrow |4a - 12| = 20$$

$$\Rightarrow 4a - 12 = \pm 20$$

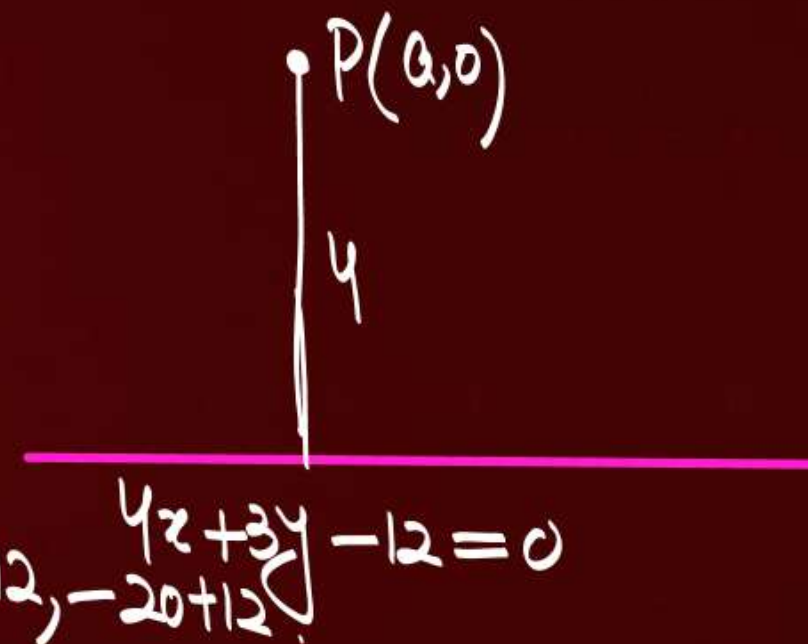
$$\Rightarrow 4a = \pm 20 + 12 = 20 + 12, -20 + 12$$

$$\Rightarrow 4a = 32, -8$$

$$\Rightarrow a = 8, -2$$

So, the required points are $(8, 0)$ & $(-2, 0)$

Hence the result.



QUESTION

Find equation of the line parallel to the line $3x - 4y + 2 = 0$ and passing through the point $(-2, 3)$.

HP/W

QUESTION

Find equation of the line perpendicular to the line $x - 7y + 5 = 0$ and having x intercept 3.

H/w.

QUESTION



Find angles between the lines $\sqrt{3}x + y = 1$ and $x + \sqrt{3}y = 1$.

(PYQJKBSE) #.

QUESTION

The line through the points $(h, 3)$ and $(4, 1)$ intersects the line $7x - 9y - 19 = 0$ at right angle. Find the value of h .

h/w

QUESTION



Prove that the line through the point (x_1, y_1) and parallel to the line $Ax + By + C = 0$ is

$$A(x - x_1) + B(y - y_1) = 0$$

h/w

QUESTION

Two lines passing through the point $(2, 3)$ intersect each other at an angle of 60° . If slope of one line is 2, find equation of the other line.

h/w

QUESTION

Find the equation of the right bisector of the line segment joining the points $(3, 4)$ and $(-1, 2)$.

H/W

QUESTION

Find the coordinates of the foot of perpendicular from the point $(-1, 3)$ to the line $3x - 4y - 16 = 0$. (Most imp)

H/W

QUESTION

The perpendicular from the origin to the line $y = mx + c$ meets it at the point $(-1, 2)$.
Find the values of m and c .

h/w

QUESTION

If p and q are the lengths of perpendiculars from the origin to the lines $x \cos \theta - y \sin \theta = k \cos 2\theta$ and $x \sec \theta + y \operatorname{cosec} \theta = k$, respectively, prove that $p^2 + 4q^2 = k^2$.

H/W

QUESTION

In the triangle ABC with vertices A (2, 3), B(4, - 1) and C(1, 2) find the equation and length of altitude from the vertex A.

H/w

QUESTION



If p is the length of perpendicular from the origin to the line whose intercepts on the axes are a and b , then show that $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$.

(PYQ JKBOSE)

$$p = \frac{|b(0) + a(0) - ab|}{\sqrt{b^2 + a^2}}$$

$$\Rightarrow p = \frac{|ab|}{\sqrt{a^2 + b^2}}$$

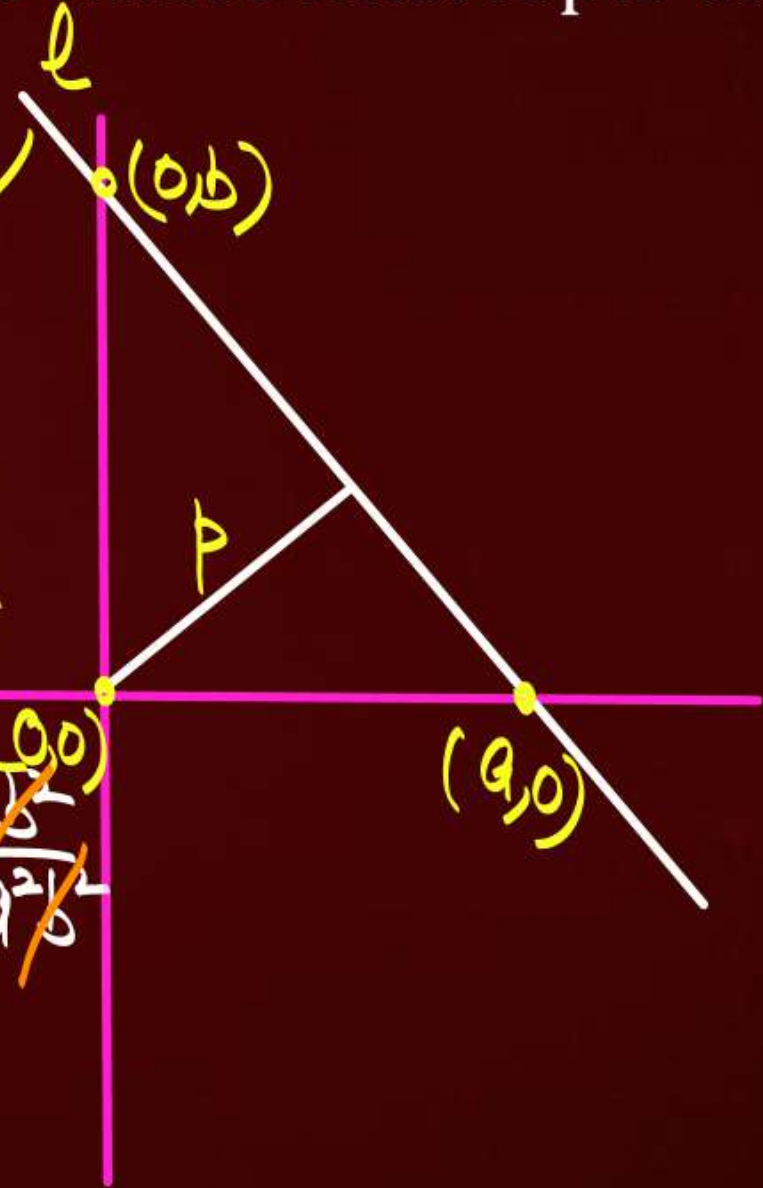
$$\Rightarrow p^2 = \frac{a^2 b^2}{\cancel{\sqrt{a^2 + b^2}}^2} = \frac{a^2 b^2}{a^2 + b^2}$$

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\Rightarrow bx + ay - ab = 0$$

$$\frac{1}{p^2} = \frac{a^2 + b^2}{a^2 b^2} = \frac{a^2}{a^2 b^2} + \frac{b^2}{a^2 b^2}$$

$$\Rightarrow \boxed{\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}}$$



Thank
You